Shaurya Varshney

+91 88515 96715 | shaurya23@iiserb.ac.in | linkedin.com/in/shaurya-varshney | github.com/shaurya3015

Summary

Computer science engineering student with hands-on experience deploying deep learning and computer vision systems to production. Built and shipped a real-time ADAS object detection pipeline (94.2% mAP) and an end-to-end RAG-based NLP engine processing 5,000+ documents. Experienced with the full ML lifecycle — model training, inference optimization, and scalable backend deployment. Seeking an AI/ML internship to apply research skills to high-impact problems.

Education

Indian Institute of Science Education and Research (IISER) — Bhopal, India
B.S. Engineering Sciences — Computer Science | CPI: 7.31/10.0 — *2023 – 2027*

- Coursework: Machine Learning, Deep Learning, Computer Vision, Intelligent Robotics, Multi-Agent RL, Data Structures & Algorithms

Experience

HCLTech — Noida, India
Software Engineering Intern — *May 2026 – Aug 2026*

- Architected a fault-tolerant telemetry ingestion pipeline processing **50M+ daily events** across distributed microservices using Vector, Docker, and ClickHouse.
- Eliminated legacy Logstash bottlenecks by deploying optimized Vector agents, reducing edge node CPU and memory utilization by **40%**.
- Engineered ClickHouse data sinks cutting storage by **65%** and analytical query latency to **¡150ms**; built Node.js schema validation services at **12,000+ req/sec**.

Transorg Analytics — Gurugram, India
Generative AI Intern — *May 2025 – Jul 2025*

- Developed an end-to-end RAG pipeline and NLP knowledge graph engine to parse **5,000+ unstructured technical profiles** using LangChain, FAISS, and transformer-based embeddings.
- Improved semantic retrieval accuracy (Hit Rate@5) by **27%** via custom embedding fine-tuning, recursive parent-child chunking, and prompt engineering.
- Optimized LLM inference latency by **30%** through hardware-accelerated local Ollama deployment, exposed via async FastAPI microservices.

Projects

Object Detection & Multi-Modal Distance Estimation — ADAS | *Python, YOLOv11, OpenCV, Kalman Filter* — 2025

- Trained and deployed a fine-tuned YOLOv11 deep learning model achieving **94.2% mAP@0.5** for multi-class object detection in a real-time computer vision pipeline at **60+ FPS**.
- Designed a monocular depth estimation module fusing pinhole camera geometry and ground-plane homography, reducing distance estimation error to **¡4.8%** over 50m without stereo hardware.
- Implemented Kalman Filter-based multi-object tracking for collision-risk inference, reducing identity switches by **35%** and enabling reliable real-time safety zone classification.

Technical Skills

Languages: Python, C++, SQL, JavaScript (ES6+), MATLAB
AI / ML: PyTorch, Scikit-Learn, YOLOv11, OpenCV, LangChain, RAG, FAISS, Ollama, Transformers, NumPy, Pandas
Backend & Data: FastAPI, Node.js, PostgreSQL, ClickHouse, Neo4j
Infrastructure: Git, Docker, Vector, Linux

Leadership

Co-Founder, Green Lungs Universal Foundation — Scaled environmental NGO across 3 municipalities with 120+ volunteers; built digital operations ledger improving allocation efficiency by **22%**.